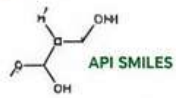
AI-DRIVEN EXCIPIENT SELECTION & FORMULATION DESIGN PIPELINE
(WITHOUT QUANTITY PREDICTION)

01 INPUT



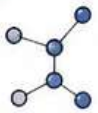
API SMILES

Route of Administration (Oral / IV / Topical ...)

Dosage / Strength (mg or mg/kg) → Defines the problem context for formulation design.

Why? Provides essential constraints and context for the entire pipeline.

02 STRUCTURAL ENCODING (MPNN)



MPNN (API) → h_api_struct

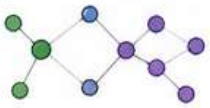
MPNN (Excipients) → h_exc_struct

Gating + Metedata

- Physicochemical
- Functional Groups
- Class / Role
- Other Meta

Why? Captures rich structural and physicochemical representations of API & excipients.

03 GRAPH ENCODING (HGT)



- API
- Excipient (E_1)
- Excipient (E_2)

Heterogeneous Graph Transformer (API – EXC – EXC) → h_api_graph, h_exc_graph

Why? Models higher-order relationships between API–Excipients and Excipient–Excipient.

04 FUSION (GATED FUSION)

h_api_struct + h_api_graph → ⊕ → h_api

h_exc_struct + h_exc_graph → ⊕ → h_exc

Why? Combines complementary information from structural and relational views via learnable gating.

05 RETRIEVAL (FAISS)

Excipients Vector Index (FAISS) → Query with h_api (+ context) → Top-K Candidate Excipients

Why? Efficiently retrieves the most relevant excipients from a large database.

06 INTERACTION MODELING

h_api + h_exc (Top-K) → Interaction

$h_{int} = h_{api} \odot h_{exc}$

FiLM Conditioning (with context) → h_out

Why? Captures fine-grained API–Excipient interaction modulated by formulation context (e.g., route).

07 SCORING (PAIR SCORER)

Concatenate: h_out, h_api, h_exc, Role Embedding → MLP (Score Network) → Compatibility Score (0 to 1)

Why? Estimates pairwise compatibility score for each API–Excipient candidate.

08 STAGE 2 SEARCH (COMBINATION GENERATION)

Top-K Excipients from Retrieval → Search / Optimization (Beam Search or GFlowNet) → Promising Excipients Combinations

Why? Finds optimal or high-quality combinations beyond greedy selection.

09 BUNDLE MODEL (SET TRANSFORMER)

Selected Combination (Set of Excipients) → Set Transformer (Encoder) → Formulation Embedding → Bundle Score (0 to 1)

Why? Evaluates the overall formulation (set-level) synergy and compatibility.

10 RULE ENGINE (HARD FILTERS + SOFT PENALTIES)

Hard Filters (Non-Negotiable)
- Regulatory Limits
- Incompatibilities
- Safety / Toxicity

Soft Penalties (Weighted)
- Stability Risk
- Solubility Risk
- Manufacturability

→ Feasible & Ranked Formulations

Why? Ensures safety, regulatory compliance and practical feasibility.

11 OUTPUT

Top-N Formulations

- Compatibility / Bundle Score
- Risk Flags (Stability, Safety, etc.)
- Recommended Amount Ranges*

→ Final Output (Without Quantity Prediction)

*Amount ranges (optional) can be suggested from rules or external databases, not predicted by the model.

DATA SOURCES

- Internal Feature DB (Features + FG Tables)
- Excipient Database (~3000+ Excipients)
- Incompatibility Sources
  - claudeIncompat.csv
  - handbookIncompat
  - PharmDE Incompat JSON
- Literature / Formulation Data (Positive Pairs)

SCORING DETAILS (PAIR SCORER)

Positive Examples (Label = 1)
- Known Compatible Pairs from Formulation Data / Literature

Incompatibility Sources (Not Hard 0)
- claudeIncompat.csv
- handbookIncompat
- PharmDE Incompat JSON

Soft Label Generation
- Analyze reaction type, severity, conditions
- Assign score in [0, 1]
  - 0.8 – 1.0 : Compatible
  - 0.4 – 0.7 : Partially Compatible
  - 0.0 – 0.3 : Strongly Incompatible

Hard Incompatibility = 0
(e.g., known severe reactions)

SCORING & RANKING OUTPUTS

- Stability Prediction
- Bioavailability Prediction
- Manufacturability Prediction
- Overall Quality Score
- Rank Order
- Explanations (Key Factors)

KEY HYPERPARAMETERS (SUMMARY)

- D-MPNN Hidden Dim: 256
- D-MPNN Depth: 4
- HGT Heads: 8, Layers: 2
- FAISS Index: IVF-PQ
- FAISS nprobe: 32
- Top-K Retrieval: 50 / 100
- Pair MLP Hidden: 256, 128
- Dropout: 0.1 – 0.2
- Set Transformer Layers: 2
- Heads: 4, Dim: 256
- Learning Rate: 1e-4
- Batch Size: 64
- Optimizer: AdamW

Input · Encoding · Retrieval · Interaction & Scoring · Search · Bundle Modeling · Filtering · Output · Why Needed?